

Ridge and Principal Component Regression:

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LSM

Model for standard linear regression

$$Y = X\beta + \varepsilon$$

$n \times 1 \quad n \times p \quad p \times 1 \quad n \times 1$

$$E\varepsilon = 0 \\ V(\varepsilon) = \sigma^2 I$$

Several "tricky" situation

- (1) $p > n$
(2) $\text{rank}(X^T X) < p$
- } no unique LSE for β

lack of identifiability of β

inflation of variance of some $\hat{\beta}_j$

$$\text{Var} \hat{\beta} = \sigma^2 (X^T X)^{-1}$$

- (3) $X^T X$ is nearly singular (i.e., there is multicollinearity)

Both ridge and PC regression may be viewed as dimension reduction techniques.

Suppose $X^T X$ is either singular or nearly singular

$$\hat{\beta} = (X^T X)^{-} X^T Y \leftarrow \text{a least squares estimate}$$

We may specify the g-inverse to be the

Moore-Penrose, i.e. $\hat{\beta} = (X^T X)^+ X^T Y$

For any parameter θ and its estimator $\hat{\theta}$, we can measure the accuracy of $\hat{\theta}$ through its MSE defined as

$$\begin{aligned} E \|\theta - \hat{\theta}\|^2 &= E \|(\hat{\theta} - E\hat{\theta}) + (E\hat{\theta} - \theta)\|^2 \\ &= \text{tr}(\text{Var} \hat{\theta}) + \underbrace{\|E\hat{\theta} - \theta\|^2}_{\text{Bias } \theta} \end{aligned}$$

Idea: We may introduce certain bias in the estimation of β to reduce its variability

Idea behind ridge regression

Introduce a dampening factor so that the variability of the estimate decreases

$$\left[\begin{array}{l} \text{SVD: } X = U \Delta^{1/2} V^T \\ \quad \quad \quad n \times p \quad p \times p \quad p \times p \\ X^T X = V \Delta V^T \quad U^T U = V^T V = I_p = W^T \\ \quad \quad \quad \Delta = \text{diag}(\delta_1, \dots, \delta_p) \end{array} \right]$$

$$\begin{aligned} \hat{\beta} &= (V \Delta V^T)^{-1} V \Delta^{1/2} (U^T Y) \quad \text{where } U^T Y = \tilde{Y} \\ &= V \Delta^{-1} [V^T Y] \Delta^{1/2} \tilde{Y} \quad Y \sim (X\beta, \sigma^2 I) \\ &= V \tilde{\Delta}^{-1/2} \tilde{Y} \end{aligned}$$

$$\begin{aligned} E \tilde{Y} &= U^T X \beta = \Delta^{1/2} V^T \beta = \Delta^{1/2} \gamma \quad \gamma = V^T \beta \\ \text{Var } \tilde{Y} &= U^T \text{Var}(Y) U = \sigma^2 U^T U = \sigma^2 I_p \end{aligned}$$

Ridge regression

$$\begin{aligned} \hat{\beta}(\lambda) &= (X^T X + \lambda I)^{-1} X^T Y \\ &= V(\Delta + \lambda I_p)^{-1} V^T V \Delta^{1/2} \tilde{Y} \\ &= V(\Delta + \lambda I_p)^{-1} \Delta^{1/2} \tilde{Y} \end{aligned}$$

$(X^T X + \lambda I)^{-1}$ exists for all $\lambda > 0$
 $\rightarrow$ Biased est of β for all $\lambda > 0$

(λ = ridge regularization parameter)

Different implication for β and $\mu = X\beta = EY$

Mean Squared error of $\hat{\beta}(\lambda)$ as estimate of β

$$\text{Let } D_{\beta}(\lambda) = E \|\hat{\beta}(\lambda) - \beta\|^2$$

Let $\hat{\mu}(\lambda) = X\hat{\beta}(\lambda) \leftarrow$ ridge regression estimate of μ

$$\text{Then } D_{\mu}(\lambda) = E \|\hat{\mu}(\lambda) - \mu\|^2 = \text{MSE of } \hat{\mu}(\lambda) \quad (\text{as est. of } \mu)$$

$$D_{\beta}(\lambda) = E \|\hat{\beta}(\lambda) - \beta\|^2 = \text{tr}(\text{Var } \hat{\beta}(\lambda)) + \|E \hat{\beta}(\lambda) - \beta\|^2$$

$$\text{Let } A(\lambda) = X^T X + \lambda I \quad \text{so } \hat{\beta}(\lambda) = (A(\lambda))^{-1} X^T Y$$

$$\text{Var } \hat{\beta}(\lambda) = \sigma^2 (A(\lambda))^{-1} X^T X (A(\lambda))^{-1}$$

$$= \sigma^2 (A(\lambda))^{-2} X^T X \quad (\text{same eigenbasis of symm. matrix, so commute})$$

$$\text{Bias } \hat{\beta}(\lambda) = E \hat{\beta}(\lambda) - \beta$$

$$= A(\lambda)^{-1} X^T X \beta - \beta$$

$$= -\lambda A(\lambda)^{-1} \beta = -\lambda V(A + \lambda I)^{-1} \underbrace{V^T \beta}_{\beta}$$

$$\begin{aligned} \text{Var } \hat{\beta}(\lambda) &= \sigma^2 (A(\lambda))^{-2} X^T X \\ &= \sigma^2 V \underbrace{[(A + \lambda I)^{-2} \Delta]}_{\substack{\delta_j \\ \frac{\delta_j}{(\delta_j + \lambda)^2} < \frac{1}{\delta_j} \quad \forall \lambda > 0}} V^T \end{aligned}$$

$\rightarrow \text{diag}\left(\frac{\delta_j}{(\delta_j + \lambda)^2}\right)$

$$\text{tr Var } \hat{\beta}(\lambda) = \sigma^2 \sum_{j=1}^p \frac{\delta_j}{(\delta_j + \lambda)^2}$$

$$\begin{aligned}\|\hat{\beta}(\lambda) - \beta\|^2 &= \|\text{Bias } \hat{\beta}(\lambda)\|^2 = \lambda^2 \gamma^T (\Delta + \lambda I)^{-2} \gamma \\ &= \sum_{j=1}^p \frac{\lambda^2}{(\delta_j + \lambda)^2} \gamma_j^2\end{aligned}$$

Combining, we have $\rightarrow$ decreasing function of λ

$$D_{\beta}(\lambda) = \sigma^2 \sum_{j=1}^p \frac{\delta_j}{(\delta_j + \lambda)^2} + \frac{\lambda^2}{(\delta_j + \lambda)^2} \gamma_j^2 \rightarrow \text{increasing function of } \lambda$$

$$\begin{aligned}\text{Expression for } D_{\mu}(\lambda) &= \mathbb{E} \|\hat{\mu}(\lambda) - \mu\|^2 \text{ where } \mu = X\beta \\ &= \mathbb{E} \|X(\hat{\beta}(\lambda) - \beta)\|^2\end{aligned}$$

$$\hat{\mu}(\lambda) = X(X^T X + \lambda I)^{-1} X^T Y = H(\lambda) Y$$

$$\text{Var } \hat{\mu}(\lambda) = \sigma^2 (H(\lambda))^2 \quad \uparrow \text{ hat matrix for ridge regression}$$

$$\begin{aligned}\text{Bias } \hat{\mu}(\lambda) &= \mathbb{E} \hat{\mu}(\lambda) - \mu = H(\lambda) X \beta - X \beta \\ &= (H(\lambda) - I) X \beta\end{aligned}$$

$$X = U \Delta^{1/2} V^T \Rightarrow X \beta = U \Delta^{1/2} \underbrace{V^T \beta}_{\gamma} = U \underbrace{\Delta^{1/2} \gamma}_{\theta} = U \theta$$

$$\Rightarrow \text{Bias } \hat{\mu}(\lambda) = (H(\lambda) - I) U \theta$$

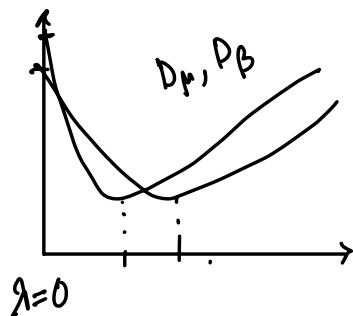
$$\left[\begin{aligned} H(\lambda) &= X(X^T X + \lambda I)^{-1} X^T \\ &= U \Delta^{1/2} \underbrace{V^T V}_{I} (\Delta + \lambda I)^{-1} \underbrace{V^T V}_{I} \Delta^{1/2} U^T \\ &= U \Delta (\Delta + \lambda I)^{-1} U^T \end{aligned} \right]$$

$$D_{\mu}(\lambda) = \sigma^2 \text{tr}(H(\lambda))^2 + \|(H(\lambda) - I)U\theta\|^2$$

Verify that,

$$D_{\mu}(\lambda) = \sigma^2 \sum \frac{\delta_j^2}{(\delta_j + \lambda)^2} + \sum \frac{\lambda^2}{(\delta_j + \lambda)^2} \theta_j^2$$

↓
decreasing function of λ ↓
increasing in
of λ



Q) Is it advantageous to use ridge regression as opposed to LS regression?

[What is the impact on $D_{\mu}(\lambda)$?]

Suppose $X^T X$ is nearly singular in the sense that there is $j_0 < p$ s.t. $\delta_{j_0+1}, \dots, \delta_p \approx 0$

First term in $D_{\mu}(\lambda)$ is then $\approx \sigma^2 \sum_{j=1}^{j_0} \left(\frac{\delta_j}{\delta_j + \lambda} \right)^2 \leq j_0 \sigma^2$

if $\lambda \gg \lambda_{j_0+1}$

$$\begin{aligned} \text{Bias component of } D_{\mu}(\lambda) &= \sum_{j=1}^p \left(1 - \frac{\delta_j}{\delta_j + \lambda} \right)^2 \theta_j^2 \\ (\theta = \Delta^{1/2} V^T \beta) &= \sum_{j=1}^{j_0} \left(1 - \frac{\delta_j}{\delta_j + \lambda} \right)^2 \theta_j^2 + \sum_{j=j_0+1}^p \left(1 - \frac{\delta_j}{\delta_j + \lambda} \right)^2 \theta_j^2 \end{aligned}$$

If $\lambda \ll \delta_{j_0}$ then bias term is approx.

$$\sum_{j=j_0+1}^p \left(1 - \frac{\delta_j}{\delta_j + \lambda} \right)^2 \theta_j^2 \leq \sum_{j=j_0+1}^p \theta_j^2$$

PC regression

$$Y = X\beta + \varepsilon$$

$$= U\theta + \varepsilon$$

↑ ↑
new new
predictor parameter

$$X = U\Delta^{1/2}V^T$$

$$\theta = \Delta^{1/2}V^T\beta$$

$$U = X\Delta^{-1/2}$$

columns of U matrix are referred to as the principal components of X

Idea of PC regression is to use first k of the PC's in the regression model $1 \leq k \leq P$

$$\hat{\theta}_j = \frac{U_{.j}^T Y}{U_{.j}^T U_{.j}} = U_{.j}^T Y$$

$$\hat{\mu}^{PC}(k) = \sum_{j=1}^k \hat{\theta}_j U_{.j}$$

$$\text{MSE for } \hat{\mu}^{PC}(k) \text{ is } D_{\mu}^{PC}(k) = E \|\hat{\mu}^{PC}(k) - \mu\|^2 \\ = k\sigma^2 + \sum_{j=k+1}^P \theta_j^2$$

$$\text{Compare with } D_{\mu}(\lambda) \approx j_0\sigma^2 + \sum_{j=j_0}^P \left(1 - \frac{\delta_j}{\delta_j + \lambda}\right)^2 \theta_j^2$$

Data driven choice of λ in ridge regression

Use a cross-validation approach

Leave i th observation (Y_i, x_i) and fit ridge regression using remaining observations

$\{(Y_j, x_j)\}_{j \neq i}$ | Prediction of Y_i based on fits from the remaining

$$Y_i = x_i^T \beta + \varepsilon_i$$

$$\hat{Y}_{\lambda,i}^{(-i)} = x_i^T \hat{\beta}^{(-i)}(\lambda)$$

$$\text{where } \hat{\beta}^{(-i)}(\lambda) = (X_{(-i)}^T X_{(-i)} + \lambda I)^{-1} X_{(-i)}^T Y_{(-i)}$$

leave one out cross validation criterion

$$CV(\lambda) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_{\lambda,i}^{(-i)})^2$$

Choose λ that minimizes this

We only need to fit ridge regression once!!

$$\text{Check that: (a) } \hat{\beta}^{(-i)}(\lambda) \stackrel{\text{def}}{=} (X_{(-i)}^T X_{(-i)} + \lambda I)^{-1} X_{(-i)}^T Y_{(-i)}$$

$$= \hat{\beta}(\lambda) - \left(\frac{Y_i - x_i^T \hat{\beta}(\lambda)}{1 - h_{ii}(\lambda)} \right) (X^T X + \lambda I)^{-1} x_i$$

where $h_{ii}(\lambda) = (i, i)^{\text{th}}$ element of $H(\lambda)$

$$= x_i^T (X^T X + \lambda I)^{-1} x_i$$

$$\text{Consequently (b) } Y_i - \hat{Y}_{\lambda,i}^{(-i)} = \frac{Y_i - x_i^T \hat{\beta}(\lambda)}{1 - h_{ii}(\lambda)}$$